
Finite-register compilation of optimal LCHS kernels: gate savings and a weight-preparation crossover

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Abstract

Optimal Fourier kernels for dissipative quantum simulation need not minimize the cost of an explicit circuit. We compile the Gaussian-damped Cauchy kernel of approximate linear combination of Hamiltonian simulations (LCHS) into a sinusoidal-grid, single-QSP selector, with verified QSVT weight preparation. An interpolation construction targets the finite register spectrum while certifying polynomial boundedness throughout the QSVT domain. On periodic advection–diffusion with $(n_x, n_k) = (6, 6)$ and $t = 0.0125$, a magnitude-QSVT compiler with counted phase lookup uses 37 389 single-target multicontrolled (STMC) gates, 25.20% fewer than the native near-optimal-kernel direct-weight baseline. Both meet $\epsilon = 10^{-3}$; the variant achieves relative error 7.77×10^{-4} , 1 104 generator queries, 15 ancillae and success probability 0.5372. With the same weight compiler for both kernels, the gate reduction is 38.86%. The explicit phase lookup costs $O(2^{n_k})$; its finite-register savings do not establish optimal asymptotic gate scaling. Longer-time $(6, 6)$ runs fail the target and are retained. A separate $(6, 9)$ study reduces gates by 24.3–33.4% relative to the direct-weight baseline using Cartesian QSVT throughout. All comparisons execute the supplied QuCF selector and sparse block encodings; every phase sequence, state vector and gate tally is retained. The results distinguish a kernel’s asymptotic cutoff advantage from its finite-circuit cost, and identify weight preparation as the main obstacle to realizing that advantage at small resolutions.

1 Introduction

The evolution $\psi(t) = e^{-At}\psi_0$ is contractive when $A = L + iH$, $L = (A + A^\dagger)/2 \succeq 0$ and $H = (A - A^\dagger)/(2i)$. Dissipation prevents direct implementation by a unitary quantum circuit. LCHS instead synthesizes the propagator from unitary evolutions generated by $H + kL$ [An et al., 2023]. Its Fourier kernel controls the largest simulated Hamiltonian, the interference normalization, and the circuit preparing the Fourier register. These costs are coupled: reducing the integration cutoff can make preparation harder or decrease the heralded success probability.

Exact LCHS kernels with stretched-exponential tails yield a cutoff $K = O(\log^{1/\beta}(1/\epsilon))$, $0 < \beta < 1$ [An et al., 2026]. Analytic extension permits approximate LCHS with $K = O(\log(1/\epsilon))$ [Low and Somma, 2025, Jin et al., 2025]. Separately, a sine coordinate turns the Fourier-index dependence into a simple block encoding and a single quantum signal processing (QSP) selector [Novikau and Joseph, 2026]. Combining these ideas requires an actual weight circuit: replacing a function in a classical integral does not establish a quantum resource improvement.

We study this combination while preserving the sine grid, qubitization walk and sparse generator block encodings. The kernel itself is the published f_2 construction, not a new analytic family. Our contribution is its verified finite-register quantum singular value transformation (QSVT) compilation and an end-to-end resource comparison at the required $(6, 6)$ registers that includes both weight preparation and the selector. We compare a Cartesian QSVT compiler with a magnitude-QSVT compiler followed by explicit diagonal phase gates; the latter uses fewer gates at this finite resolution.

A useful technical point is that a weight polynomial need only approximate the target at the register eigenvalues; it must still remain bounded between them. We separate these two requirements and certify both. The resulting experiment exposes two distinct questions: which kernel is cheaper under a common QSVT compiler, and when that compiler beats the direct controlled-rotation implementation used in earlier numerical work. All reported accuracy and quantum resource counts are obtained by circuit execution. The committed headline axis is total STMC gate complexity (axis 3). Query and workspace counts are reported as supporting resources; neither is substituted for the gate objective after observing the results.

Relation to prior work. The original LCHS construction [An et al., 2023] and its explicit advection–diffusion implementation [Novikau and Joseph, 2025] established the algorithmic and numerical setting. The exact-kernel improvement and its analyticity obstruction are due to An et al. [2026]. We adopt Low–Somma’s shifted Gaussian-damped Cauchy kernel rather than claiming its logarithmic cutoff as a new result. Smooth auxiliary initializations in Schrödingerization provide another route to optimal matrix queries [Jin et al., 2025]. Fourier signal processing [de Lima Silva et al., 2022], qubitization [Low and Chuang, 2019], and QSVT [Gilyén et al., 2019] supply the relevant function-transformation tools. Recent work extends LCHS to general analytic matrix functions using Weyl calculus [Ni and Ying, 2026] and unifies recovery procedures for time-dependent dynamics [Dong et al., 2026]. Those developments allow broader changes to the representation; here the circuit and quadrature structures are deliberately fixed. Our numerical Newton phase solver follows the symmetric-QSP formulation [Dong et al., 2021]; recent analytic angle methods [Bernard and Wiebe, 2026] are complementary, and were not imported as implementation code.

2 Kernels, quadrature and circuit compilation

2.1 An exact kernel and an approximate kernel

Our normalization is

$$e^{-At} \simeq \int_{-K}^K \rho(k) e^{-it(H+kL)} dk, \quad \rho(k) = \frac{\xi(k)}{1 - ik}. \quad (1)$$

The reference family is

$$\xi_\beta(k) = \frac{\exp[2^\beta - (1 + ik)^\beta]}{2\pi}, \quad 0 < \beta < 1. \quad (2)$$

For comparison we use the f_2 analytic-extension kernel [Low and Somma, 2025, Eq. (53)], converted to Eq. (1):

$$\begin{aligned} \rho_{\gamma,c}(k) &= \frac{\exp[c - ick - (k^2 + 1)/(4\gamma^2)]}{\pi(1 + k^2)}, \\ \xi_{\gamma,c}(k) &= \frac{\exp[c - ick - (k^2 + 1)/(4\gamma^2)]}{\pi(1 + ik)}. \end{aligned} \quad (3)$$

Here $c > 0$ sets the analytic-extension shift and $\gamma > 0$ controls smoothing. Its total variation is $e^c \operatorname{erfc}(1/(2\gamma))$, so increasing c trades a shorter cutoff against reduced success. The lower-half-plane residue is correct, but the Gaussian does not decay along an infinitely deep contour. We therefore use approximate, not exact, admissibility. A finite-contour bound and an independent residue/integral check appear in Appendix A. At fixed c , choosing $\gamma = \Theta(\sqrt{\log(1/\epsilon)})$ gives $K = \Theta(\log(1/\epsilon))$. The exact kernel instead has the stretched-exponential cutoff in Eq. (2).

2.2 The fixed selector and discrete weights

For $N = 2^{n_k}$, retain exactly the implemented sine-grid rule

$$\theta_j = -\frac{\pi}{2} + \frac{\pi j}{N-1}, \quad k_j = K \sin \theta_j, \quad w_j = K \cos \theta_j \frac{\pi}{N-1} \rho(k_j). \quad (4)$$

The endpoint weights vanish. These Chebyshev-extremum nodes are the FCC-type discretization of the reference implementation. Equation (4) specifies the actual weights used; no additional classical

Clenshaw–Curtis weights or alternative quadrature rule are substituted. The distinction between node placement and integration weights matters when comparing quadrature formulas [Trefethen, 2008].

The unchanged QuCF routine `qsp_ham_opt2` implements $\sum_j |j\rangle\langle j| \otimes e^{-it(H+k_j L)}$ with the reference qubitization operators. Its sparse ADE oracles explicitly execute their controlled shifts and rotations. If B is their common normalization, the combined block is $(H + k_j L)/(2B)$. We choose $B \geq \max(8KD/\Delta x^2, v/\Delta x)$ and use supplied high-precision Hamiltonian-QSP phases, adjusting B so that their repeat count gives the requested time exactly. Both families use the same rule for choosing the lowest-query supplied phase file. Repetition is the reference compiler’s repetition of a single register-controlled QSP circuit, not a separate evolution for each Fourier node.

2.3 Finite-register QSVT weights

Let $x_j = \sin \theta_j$ and choose the continuous branch $q_j^2 = w_j$. For example, the new kernel has

$$q(x) = \sqrt{\frac{K}{N-1}} \frac{(1-x^2)^{1/4}}{\sqrt{1+K^2 x^2}} \exp \left[\frac{c - icKx}{2} - \frac{K^2 x^2 + 1}{8\gamma^2} \right]. \quad (5)$$

Cartesian compilation. The real part is even and the imaginary part is odd. We interpolate their node values with a discrete cosine transform, then truncate parity-preserving coefficients until each component’s discarded absolute coefficient sum is below $\epsilon/1000$. The combined complex node error is consequently bounded by $\sqrt{2}\epsilon/1000$, up to roundoff. The resulting degree is at most $N-1$. This is a finite-spectrum construction: the fractional endpoint factor in Eq. (5) can prevent equally rapid uniform approximation of the continuum function. We log both discrepancies, without treating them as interchangeable.

The two parity polynomials are rescaled, synthesized with symmetric Newton QSP iterations, and checked independently on a dense grid. A derivative bound certifies their global amplitude constraint (Appendix A). The existing QuCF parity compiler contributes a phase i^d for degree d ; explicit branch phase gates correct it before coherent real/imaginary combination. Three weight ancillae suffice: the sine block, the QSVT projection and the parity branch. We choose the normalization to make an integer number of Grover iterations reach unit ideal preparation success, and reuse this workspace for the conjugate preparation after the selector. All amplification gates are included. The resulting zero-ancilla block is $\sum_j \tilde{q}_j^2 e^{-it(H+k_j L)} / \tilde{\alpha}$, where $\tilde{\alpha} = \sum_j |\tilde{q}_j|^2$ is known from the compiled polynomials.

Polar compilation at the required registers. Write $q_j = a_j e^{i\phi_j}$ with $a_j = |q_j|$. A single even polynomial interpolates a_j with the same node budget and global boundedness check. Its QSVT circuit uses two weight ancillae and has raw preparation success $\tilde{\alpha}/(Ns^2)$, where $\tilde{\alpha} = \sum_j \tilde{a}_j^2$ and s is the polynomial scale. Choose the smallest integer r for which $s = \sqrt{\tilde{\alpha}/N} / \sin[\pi/(4r+2)]$ satisfies the amplitude constraint. Grover amplification then gives a unit-success ideal preparation $P|0\rangle = \sum_j \tilde{a}_j |j\rangle / \sqrt{\tilde{\alpha}}$, up to a known global sign. The diagonal $D_\phi = \sum_j e^{i\phi_j} |j\rangle\langle j|$ is compiled as an explicit controlled-Phase gate for each node, with counted X gates when the target bit is zero. Executing

$$(P^\dagger D_\phi) \text{ SELECT } (D_\phi P) \quad \mapsto \quad \frac{1}{\tilde{\alpha}} \sum_j \tilde{a}_j^2 e^{2i\phi_j} V_j \quad (6)$$

in the zero-ancilla block implements the same complex LCHS weights. The phase layers occur after amplification and before its inverse; their gates are all counted. This is a finite-register hybrid with $O(N)$ explicit phase gates, rather than a purely polynomial phase oracle. Both kernel families use this identical compiler. We retain the Cartesian implementation and the native direct-weight control to distinguish a kernel benefit from a compiler benefit.

3 Experimental design and resource accounting

We discretize $\partial_t \psi = -v\partial_x \psi + D\partial_x^2 \psi$ with $v = 1$, $D = 0.01$, centered differences and cyclic boundary indices. As in the supplied ADE implementation, $\Delta x = (2^{n_x} - 1)^{-1}$; the Gaussian is proportional to $\exp[-(x-0.5)^2/(2(0.05)^2)]$. This reproduces the reference discrete problem rather

than claiming a continuum-converged PDE calculation. The normalized Gaussian is prepared by an explicit binary tree of controlled R_y gates. This common initialization replaces the reference rejection sampler for both kernels and the matched direct-weight controls, and its cost is counted.

The required headline is $(n_x, n_k, t, \epsilon) = (6, 6, 0.0125, 10^{-3})$, with $\|At\|_2 = 1.9845$. Its near-optimal baseline uses $(\beta, K) = (0.745, 20.75)$ and the variant uses $(K, c, \gamma) = (6.75, 0.5, 3)$. The physical time is short because a 64-node Fourier register restricts resolvable cutoffs; the norm-time product is not negligible. Both kernels and all compilers use the same spatial generator, initial state, time and error target.

Design chronology and calibration. The original core study used $(4, 7, 0.1, 10^{-3})$ as its reference point and $\beta = 0.7$ throughout. It omitted the required $(6, 6)$ headline; the present revision adds that study and preserves the original data. The core precision targets are 0.03, 0.01, 0.003, and 0.001 at the same registers and time. Resolution tests add $(n_k, \epsilon) = (5, 0.01)$ and $(6, 0.003)$; a time sweep adds $t = 0.05, 0.2$ at the core registers and target. A timestamped plan precedes all validation. Classical calibration considers the full declared c, γ, K grid and minimizes a fixed cost proxy subject to error and success constraints. The core reference choices are $K = 32$ for the exact kernel and $(K, c, \gamma) = (18, 0.35, 6)$ for f_2 . The calibration requirement was tightened to demand a smaller variant error before any circuit validation; both versions and every candidate remain available. If a coarse baseline has no feasible candidate, its minimum-error candidate is still executed and reported.

After the core results, we declared an additional $(n_x, n_k) = (6, 9)$ experiment at $t = 0.4, 0.6, 0.8, \epsilon = 10^{-3}$, inheriting those core kernel parameters without retuning. We also declared direct controlled-rotation β controls at all matched settings. This extension is identified separately throughout. Its purpose is to test the cost of the published numerical weight implementation, which is direct preparation rather than QSVT.

For the added $(6, 6)$ study, an initial $\beta = 0.7$ calibration failed to reach 10^{-3} over the declared times. We widened the search to $\beta \in [0.7, 0.8]$, refined K to increments of 0.25, and checked local continuous (β, K) refinement. Before any new circuit run, we relaxed the diagnostic margin from 0.8ϵ to 0.95ϵ and froze the largest passing declared time, $t = 0.0125$, plus $t = 0.003125, 0.00625$ and a failed $t = 0.1$ control. The variant must also beat the paired diagnostic error and success under the original relative constraints. Every calibration candidate and refinement is retained. Baseline parameters may vary across these short-time points; they do not constitute a fixed-kernel asymptotic time test. At $t = 0.1$ we execute the original $\beta = 0.7$ diagnostic minimizer and its paired f_2 candidate, including their target failures. After the Cartesian circuit results, a separate timestamped plan introduces polar compilation at all eight frozen kernel settings, without retuning. These are disclosed successive design decisions, not an initial preregistration of the final headline.

All state vectors use the provided QuCF–QuEST GPU backend [Jones et al., 2019], double precision and one NVIDIA H200. There are no shot estimates or variational replacements. The measured physical state is the raw zero-ancilla output multiplied by the known LCU/Grover normalization. Its error is $\|\psi_{\text{circuit}} - \exp[-(L + iH)t]\psi_0\|_2 / \|\exp[-(L + iH)t]\psi_0\|_2$, with the matrix exponential evaluated at runtime. No phase alignment or reference-dependent rescaling is performed. Success is the raw probability of zero on all heralding ancillae, including the Fourier register.

STMC counts are executed flat circuit lengths, independently tallied and including initialization, all block-encoding gates, and amplification. Markers count controlled and adjoint calls to both generator oracles, $Q_A = Q_H + Q_L$. Cartesian QSVT and native direct controls allocate $n_k + 10$ ancillae: n_k Fourier-index qubits, three for weights, two for QSP, and five for the combined block encoding. Polar compilation uses two weight ancillae, hence $n_k + 9$ total. We count allocated workspace even when a direct-weight subroutine could be compiled with fewer qubits; no memory-axis advantage is claimed.

4 Results

4.1 The required (6,6) headline

Table 1 begins with the required $(n_x, n_k, t, \epsilon) = (6, 6, 0.0125, 10^{-3})$ comparison. The native Novikau–Joseph direct-weight baseline uses 49 982 STMC gates and achieves relative error 9.19×10^{-4} , 1 952 generator queries, 16 ancillae and success 0.4947. The f_2 polar circuit uses 37 389 gates,

Table 1: Required (6, 6) headline (first three rows), followed by the original core precision and resolution study. NJ direct is a newly executed near-optimal β baseline using the native Novikau–Joseph weight path. Polar and Cart. denote magnitude-QSVT/phase lookup and Cartesian QSVT, respectively. All methods use the same selector, quadrature and explicit block encodings. Ancillae include n_k ; $\dagger$ denotes a missed accuracy target.

Method	Test	(n_x, n_k, t, ϵ)	Rel. error	STMC	Q_A	Anc.	p_{succ}
NJ direct	ADE	$(6, 6, 0.0125, 1 \times 10^{-3})$	9.19×10^{-4}	49 982	1 952	16	0.4947
β polar	ADE	$(6, 6, 0.0125, 1 \times 10^{-3})$	9.19×10^{-4}	61 152	1 952	15	0.4975
f_2 polar	ADE	$(6, 6, 0.0125, 1 \times 10^{-3})$	7.77×10^{-4}	37 389	1 104	15	0.5372
β Cart.	ADE	$(4, 7, 0.1, 1 \times 10^{-3})$	5.22×10^{-4}	153 623	1 800	17	0.4543
f_2 Cart.	ADE	$(4, 7, 0.1, 1 \times 10^{-3})$	4.40×10^{-4}	139 854	1 104	17	0.4670
β Cart.	ADE	$(4, 7, 0.1, 3 \times 10^{-2})$	1.25×10^{-2}	82 894	832	17	0.4881
f_2 Cart.	ADE	$(4, 7, 0.1, 3 \times 10^{-2})$	1.15×10^{-2}	75 701	416	17	0.6111
β Cart.	ADE	$(4, 7, 0.1, 1 \times 10^{-2})$	5.32×10^{-3}	114 634	1 104	17	0.4600
f_2 Cart.	ADE	$(4, 7, 0.1, 1 \times 10^{-2})$	4.30×10^{-3}	86 923	552	17	0.4772
β Cart.	ADE	$(4, 7, 0.1, 3 \times 10^{-3})$	1.70×10^{-3}	139 339	1 656	17	0.4581
f_2 Cart.	ADE	$(4, 7, 0.1, 3 \times 10^{-3})$	1.63×10^{-3}	114 544	1 104	17	0.5736
β Cart.	ADE	$(4, 5, 0.1, 1 \times 10^{-2})$	$1.31 \times 10^{-2\dagger}$	33 364	832	15	0.4867
f_2 Cart.	ADE	$(4, 5, 0.1, 1 \times 10^{-2})$	4.67×10^{-3}	22 397	416	15	0.5369
β Cart.	ADE	$(4, 6, 0.1, 3 \times 10^{-3})$	$3.89 \times 10^{-3\dagger}$	82 003	1 656	16	0.4567
f_2 Cart.	ADE	$(4, 6, 0.1, 3 \times 10^{-3})$	1.68×10^{-3}	59 298	832	16	0.5392

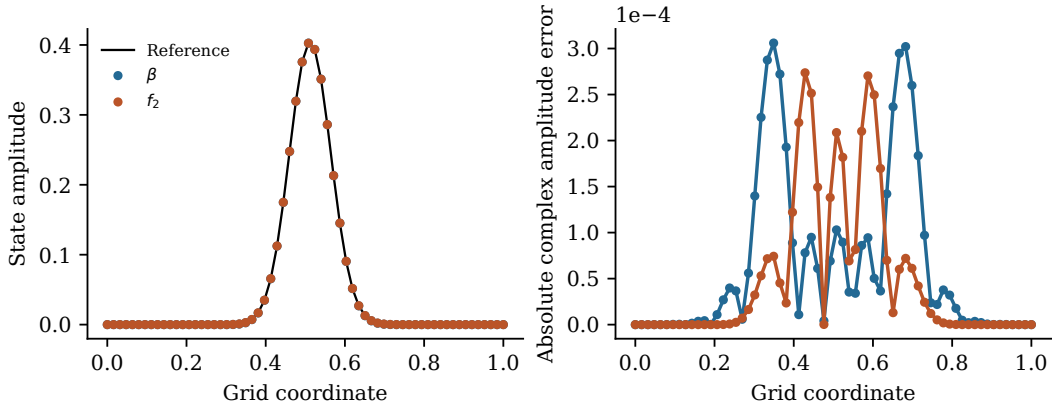


Figure 1: Required (6, 6) headline: polar-QSVT circuit outputs versus the runtime matrix exponential (left), and absolute complex amplitude errors (right), at $t = 0.0125$. The dissipative norm is retained. Error is measured against the same discrete generator, not the continuum PDE.

a 25.20% reduction, with error 7.77×10^{-4} , 1 104 queries, 15 ancillae and success 0.5372. Both errors are below the same 10^{-3} target, and the variant’s error is smaller. This is an executed comparison against the supplied numerical weight path, with all block-encoding and preparation costs included.

The common polar compiler isolates the kernel effect: its exact-kernel baseline costs 61 152 gates, so the kernel substitution alone saves 38.86%. Both polar circuits use 15 ancillae, and the baseline success is 0.4975. The Cartesian compiler also favors f_2 by 29.41%, but its variant is more expensive than the native direct baseline. These results distinguish the measured kernel benefit from the additional benefit of polar compilation. Figure 1 compares physical, unnormalized states; no output shape is renormalized or phase-aligned to the reference.

At the three target-passing (6, 6) times, f_2 polar uses fewer gates than β polar throughout. It also beats native direct weights at $t = 0.00625$ and 0.0125 , but loses at $t = 0.003125$. Figure 2 and Table 2 report $G = a\|At\|_2^p$ fits: the polar exponents are 0.683 and 0.439, while the direct baseline gives 0.852. The baseline kernel parameters vary with time under the calibration rule; these are optimized finite-setting cost curves, not a fixed-kernel asymptotic scaling law. All five compiler/kernel choices at all four times are reported in Table 6.

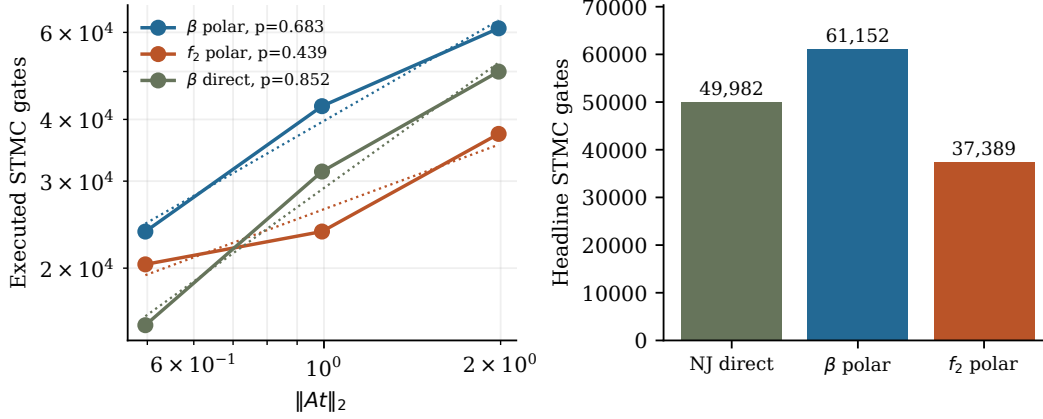


Figure 2: Required-register cost study at $\epsilon = 10^{-3}$. Left: three passing times at $(6, 6)$, with dotted descriptive fits; all errors are below target. The shortest-time direct baseline is cheaper than f_2 polar. Right: total gates at the headline $t = 0.0125$. Polar preparation uses an explicit $O(2^{n_k})$ phase lookup, whose cost is included in every plotted count.

Table 2: Descriptive fits $G = ax^p$ to total executed STMC counts. Precision uses $x = \log(1/\epsilon)$; time uses $x = \|At\|_2$. SE is the residual standard error of the slope, not a statistical uncertainty from sampled quantum measurements.

Sweep	Kernel	Prefactor a	Exponent p	SE	R^2
Required registers, time	β polar	39727.2	0.683	0.092	0.9821
Required registers, time	f_2 polar	26323.5	0.439	0.126	0.9240
Required registers, time	β direct	29054.4	0.852	0.104	0.9853
Core precision	β Cartesian	27128.8	0.916	0.096	0.9786
Core precision	f_2 Cartesian	22850.6	0.920	0.123	0.9656
Core time	β Cartesian	143965.4	0.191	0.026	0.9821
Core time	f_2 Cartesian	132082.1	0.092	0.020	0.9566
Extended time	β Cartesian	95559.1	0.785	0.124	0.9757
Extended time	f_2 Cartesian	58229.8	0.784	0.002	1.0000

4.2 Core scaling and resolution characterization

The preserved $(4, 7, 0.1, 10^{-3})$ core reference costs 153 623 and 139 854 gates with Cartesian QSVT, a 8.96% reduction. Its precision fit $G = a[\log(1/\epsilon)]^p$ gives exponents 0.916 and 0.920, with residual standard errors 0.096 and 0.123. This sweep does not establish a lower precision exponent or prove optimal end-to-end logarithmic scaling. The independent core time fit gives $p = 0.191$ and $p = 0.092$; the variant’s measured exponent is lower. Figure 3 and Tables 7 and 2 retain all points, prefactors and exponents. These deterministic-count fit uncertainties summarize curvature about a power law, not quantum measurement noise.

At $(n_k, \epsilon) = (5, 0.01)$ and $(6, 0.003)$ with $n_x = 4$, the best calibrated exact-kernel candidates miss their targets; both f_2 circuits pass (Fig. 4). These pairs demonstrate a larger feasible resolution range, not matched-accuracy speedups. At $(7, 0.001)$ both kernels pass. Thus f_2 is preferable among the two families under the evaluated common QSVT compiler across these three regimes, but direct preparation can change the practical winner. This is not a global optimization over all analytic-extension kernels.

4.3 The implementation crossover and longer-time failures

The native direct control costs only 42 416 gates at the $(4, 7)$ core reference, below either Cartesian-QSVT circuit. Table 8 reports all core direct controls. The separately declared $(6, 9)$ extension inherits fixed kernel parameters from that core point. Across $t = 0.4, 0.6, 0.8$, Cartesian f_2 saves 36.1–41.1% against Cartesian β , and 24.3–33.4% against native direct β . All three variants have

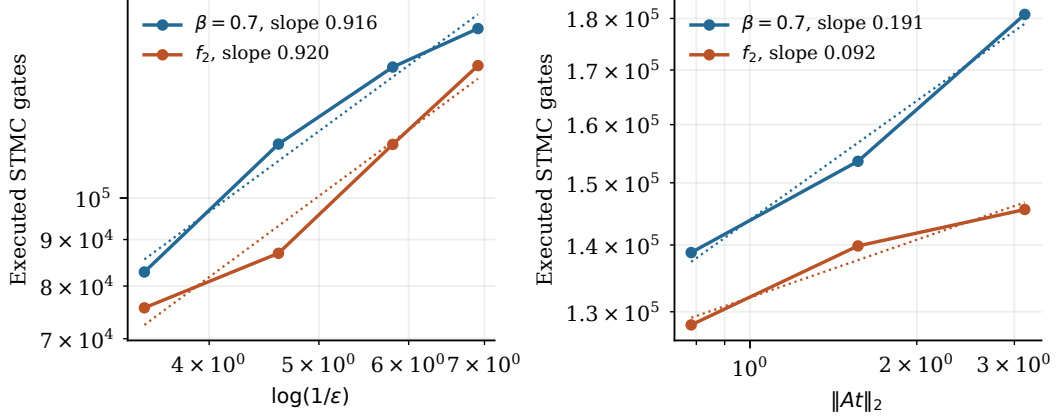


Figure 3: Executed Cartesian-QSVT total STMC counts for the core (4, 7) precision sweep (left) and time sweep (right). Dotted curves are descriptive power-law fits. Small fitted time exponents reflect fixed preparation overhead, not fast forwarding. The precision exponents are nearly equal.

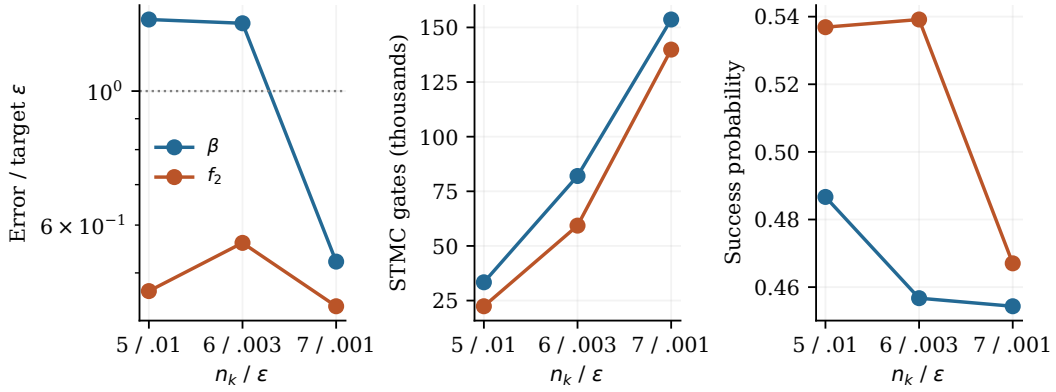


Figure 4: Kernel-family characterization from executed Cartesian-QSVT circuits at $n_x = 4$. Error divided by target exceeds one for two exact-kernel cases; both failures are retained. The other panels show executed gates and success probability.

smaller state error and higher success than both references. Their fixed-parameter time exponents, 0.785 and 0.784, are nearly equal. This supplies three matched-accuracy gate reductions without the polar phase lookup, with a prefactor improvement consistent with unchanged large-time selector scaling.

The added (6, 6, $t = 0.1$) circuits all fail the 10^{-3} target: Table 6 reports both kernels and all compilers. The classical (β , K) refinements also failed to find a passing near-optimal baseline at this longer time. This is an observed limit of the declared search and Fourier resolution, not an impossibility result for every kernel parameter. We therefore make no matched-accuracy speedup claim for this time. Switching the weight compiler preserves these quadrature errors and cannot repair them.

4.4 Classical work and independent checks

The original calibration evaluated 29 640 candidates in 11.43 seconds. Additional required-register calibration is documented separately in Appendix B. Across selected configurations, median discretization times were 60.6 and 54.0 microseconds for β and f_2 . Median Cartesian weight-synthesis times, including dense phase validation, were 0.456 and 0.442 seconds; totals were 8.693 and 4.891 seconds. Polar synthesis medians were 0.110 and 0.110 seconds, with 0.983 seconds across all six distinct polar circuits. Supplied selector phases are reused for both families; their original generation

Table 3: Declared larger-register extension with inherited, fixed kernel parameters. Direct denotes the supplied controlled-rotation weight construction; its amplification cost is included.

Method	Test	(n_x, n_k, t, ϵ)	Rel. error	STMC	Q_A	Anc.	p_{succ}
β Cart.	ADE	$(6, 9, 0.4, 1 \times 10^{-3})$	5.97×10^{-4}	2 539 835	76 104	19	0.2880
β direct	ADE	$(6, 9, 0.4, 1 \times 10^{-3})$	5.97×10^{-4}	2 030 312	76 104	19	0.2879
f_2 Cart.	ADE	$(6, 9, 0.4, 1 \times 10^{-3})$	5.07×10^{-4}	1 510 875	41 568	19	0.2957
β Cart.	ADE	$(6, 9, 0.6, 1 \times 10^{-3})$	6.11×10^{-4}	3 255 280	103 360	19	0.2450
β direct	ADE	$(6, 9, 0.6, 1 \times 10^{-3})$	6.11×10^{-4}	2 745 757	103 360	19	0.2449
f_2 Cart.	ADE	$(6, 9, 0.6, 1 \times 10^{-3})$	4.10×10^{-4}	2 079 145	63 216	19	0.2514
β Cart.	ADE	$(6, 9, 0.8, 1 \times 10^{-3})$	9.95×10^{-4}	4 414 085	147 504	19	0.2167
β direct	ADE	$(6, 9, 0.8, 1 \times 10^{-3})$	9.95×10^{-4}	3 904 562	147 504	19	0.2167
f_2 Cart.	ADE	$(6, 9, 0.8, 1 \times 10^{-3})$	5.26×10^{-4}	2 602 040	83 136	19	0.2225

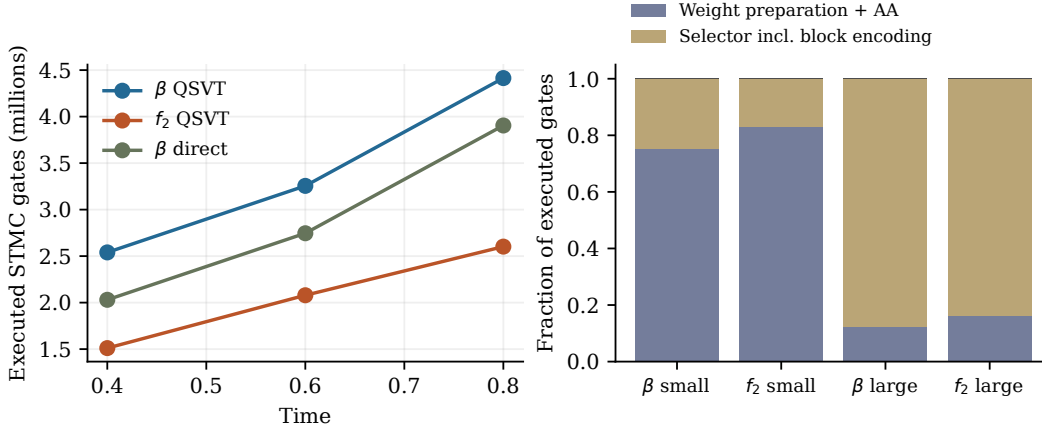


Figure 5: The larger-register extension compares Cartesian-QSVT kernels with native direct weights (left). Executed gate fractions (right) use “small” for the original $(4, 7, 0.1)$ core reference and “large” for $(6, 9, 0.8)$. Preparation includes both directions and every amplification round; initialization is the small remainder.

time is not charged as new work. Timings are single-run wall measurements, not comparative speed benchmarks.

For Cartesian weights, the largest dense-grid phase error was 1.08×10^{-11} , and standalone QuCF amplitude discrepancy before or after amplification was 1.98×10^{-12} . The maximum finite-node square-root-weight error was 2.39×10^{-5} , versus continuum discrepancy 1.54×10^{-2} ; the largest certified normalized polynomial bound was $0.824706 < 1$. For polar weights the corresponding phase, standalone and node errors were 7.06×10^{-13} , 1.44×10^{-13} and 2.22×10^{-16} , with certified bound $0.792622 < 1$. The 110 reported circuit executions took 0.85 GPU-host wall hours including construction and output. An unchanged source ADE input reproduced its archived amplitudes bit for bit, including 5 325 230 gates and success 0.2158.

5 Mechanism, failures and scope

Why the kernel helps. The selector normalization grows with $K\|L\|$ while its structure is unchanged. A Gaussian Fourier tail reaches a useful truncation with a smaller K than the stretched-exponential exact kernel. At the headline, this reduces generator queries by 43.44%. Both polar magnitude polynomials have degree 62; the variant saves selector calls through its smaller cutoff and uses three amplification rounds instead of four. Cartesian compilation retains two long parity polynomials and has larger fixed overhead. The kernel is therefore favorable because it balances cutoff, normalization and compiled preparation cost, not simply because its tail is asymptotically optimal. The retained coefficient spectra in Fig. 6 show this distinction: the new kernel’s imaginary polynomial becomes shorter at larger resolution, while both real polynomials remain long.

This distinction is also visible in the complexity accounting. With d_R, d_I parity degrees and r Grover rounds, the explicitly compiled weight cost is

$$G_w = O((2r + 1)[n_k(d_R + d_I) + d_R + d_I + n_k]), \quad (7)$$

in addition to the selector and initialization. The two preparations only multiply this expression by a constant. For resolved kernels with bounded total variation, the chosen normalization gives $r = O(\sqrt{K})$ up to interpolation-envelope factors. Thus optimal cutoff scaling does not make the complete STMC cost linear in $\log(1/\epsilon)$. Under the analytic-regime quadrature and weight approximation assumptions of Novikau and Joseph [2026, Eqs. (94)–(96)], the kernel-dependent power in the weight bound changes from $1 + 3/(2\beta)$ to $5/2$, with the same additional quadrature factors. This conditional theoretical improvement motivates the substitution; the finite-register experiment does not establish those asymptotic exponents. For polar compilation, the analogous bound is $G_w^{\text{polar}} = O((2r + 1)n_k d + 2^{n_k})$. The explicit phase lookup is useful at $n_k = 6$ but is exponential in register size; we do not transfer the Cartesian asymptotic bound to this hybrid compiler. At fixed kernel and resolution, both selectors retain linear large-time cost. The observed smaller time slopes arise from different coefficients and fixed preparation overheads.

When the advantage fails. First, direct preparation is cheaper than Cartesian QSVT on the core small instances, and cheaper than polar f_2 at the shortest $(6, 6)$ time. The new headline advantage is consequently not universal. Asymptotic QSVT state preparation is not automatically the best finite compiler. Second, increasing c can improve decay yet make $e^c \operatorname{erfc}(1/(2\gamma))$ too large: lowering circuit cost while reducing success would be a different tradeoff. The calibration explicitly rejects such candidates. Third, increasing K at fixed n_k coarsens the grid near $k = 0$ and can increase quadrature error. The retained coarse-grid failures and full candidate archive expose this nonmonotonicity. The $(6, 6, t = 0.1)$ failures persist under both weight compilers, despite accurate standalone weights. Finally, a larger approximation degree cannot repair aliasing in the LCHS sum; it only improves the weight function at the same nodes.

The branch phase errors encountered during compiler development were caught by standalone amplitude checks, despite apparently reasonable success probabilities. The rejected inputs and outputs remain in the attempts log. They were not used in any reported performance row. The Cauchy kernel and the growing-one-norm f_1 construction were considered analytically but not run as competing quantum circuits. Higher analytic-extension families may give better constants; no claim of superiority over unimplemented kernels is made.

Limitations. This is a noise-free state-vector study of a commuting, constant-coefficient discrete PDE. It neither establishes a quantum advantage over classical propagation nor tests nonnormal generators, nonlinear embeddings, time-dependent coefficients, hardware noise or a continuum limit. Calibration uses the same known generator and Gaussian input as validation; there is no held-out application claim. The extended time points inherit parameters without retuning, but remain within that same problem. The approximate kernel has a general finite-contour justification; the reported precision is state-specific, not a proved operator-norm error bound at each selected parameter tuple. State-dependent decay also limits heralded success at longer times. The required-register time and baseline exponent were calibrated after the original study; the polar compiler was introduced after the Cartesian circuit comparison. All stages are timestamped and disclosed, but this adaptivity rules out a claim of an untouched confirmatory test. The finite interpolant is certified for the executed register; extrapolating its degree scaling to much higher resolution requires new analysis. STMC counts do not include a fault-tolerant synthesis into Clifford+ T , and classical simulation time is not quantum execution time.

6 Conclusion

An optimal-cutoff LCHS kernel can be compiled into the existing sine-grid selector with explicit QSVT weights, without changing the qubitization walk or hiding the generator block encodings. At the required $(6, 6)$ registers and 10^{-3} target, magnitude QSVT with counted phase lookup saves 25.20% of gates against the native near-optimal-kernel direct baseline and 38.86% under a common compiler. The longer-time failures, finite-register phase cost and retained direct-weight controls delimit this result. Mathematical kernel improvement and practical circuit improvement depend on both resolution and the weight compiler.

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A Kernel admissibility and the finite-register error bound

We use the Fourier convention $U = \int \rho(k)V(k)dk$, without a factor $1/\sqrt{2\pi}$ outside the integral. For the exact kernel, $\xi_\beta(z) = \exp[2^\beta - (1+iz)^\beta]/(2\pi)$ is analytic for $\text{Im } z < 0$: the argument of the principal power has positive real part there. It is continuous on the boundary, decays on the real axis, and $\xi_\beta(-i) = 1/(2\pi)$. Consequently ρ_β has residue $i/(2\pi)$ at $-i$, giving the normalized propagator under a clockwise lower-half-plane contour. These are the exact-LCHS conditions of An et al. [2026].

For $\rho_{\gamma,c}$ in Eq. (3), the same residue holds, but the Gaussian grows along the negative imaginary axis. Closing an infinite semicircle is therefore invalid. Instead use the finite horizontal contour $z = u - iy$, $y = 2c\gamma^2 > 1$. Contractivity of $\exp[-it(H + zL)]$ for $L \succeq 0$ bounds the operator error by the real-axis tail plus the lower horizontal integral. The identity

$$|1 + (u - iy)^2| = \sqrt{[u^2 + (y-1)^2][u^2 + (y+1)^2]} \geq u^2 + (y-1)^2 \quad (8)$$

and a Gaussian tail estimate give the convenient bounds

$$E_{\text{contour}} \leq \frac{\exp[c - c^2\gamma^2 - 1/(4\gamma^2)]}{2c\gamma^2 - 1}, \quad (9)$$

$$E_{\text{tail}} \leq \frac{4\gamma^2 \exp[c - (K^2 + 1)/(4\gamma^2)]}{\pi K(1 + K^2)}. \quad (10)$$

Both vanish exponentially when $\gamma^2 = \Theta(\log(1/\epsilon))$ and $K = \Theta(\log(1/\epsilon))$ at fixed $c > 0$. The residue, finite contour and decay conditions thus establish approximate admissibility. This is the published Gaussian-damped Cauchy construction, with the normalization converted to our convention [Low and Somma, 2025]. The independently evaluated integrals, residue and these bounds are saved in `logs/kernel_checks.json`. Parameters calibrated to the Gaussian test state need not satisfy a uniform operator-error budget of the same numerical value; the measured state error is the reported accuracy.

Let $q_j^2 = w_j$ on the finite grid, with a continuous square-root branch, and let $\tilde{q}_j = P_R(x_j) + iP_I(x_j)$ be the compiled polynomials. For any unitary selector blocks V_j ,

$$\left\| \sum_j (\tilde{q}_j^2 - q_j^2) V_j \right\| \leq \sum_j |\tilde{q}_j - q_j| |\tilde{q}_j + q_j| \quad (11)$$

$$\leq \|\tilde{q} - q\|_2 (2\|q\|_2 + \|\tilde{q} - q\|_2) \quad (12)$$

$$\leq \sqrt{N}\delta(2\sqrt{\alpha} + \sqrt{N}\delta), \quad \alpha = \sum_j |w_j|, \quad \delta = \max_j |\tilde{q}_j - q_j|. \quad (13)$$

This bound needs accuracy only at the register eigenvalues. The interpolating polynomial still must be bounded on the entire QSVT domain. For $P(x) = \sum_{\ell=0}^d a_\ell T_\ell(x)$ and a uniform verification grid with M points, $|T'_\ell(x)| \leq \ell^2$ implies

$$\|P\|_\infty \leq \max_{x \in \mathcal{G}_M} |P(x)| + \frac{\sum_\ell \ell^2 |a_\ell|}{M-1}. \quad (14)$$

Thus a grid check with this explicit remainder certifies feasibility; a dense check alone would not do so. The machine-precision coefficient roundoff is negligible compared with the recorded feasibility margin.

After the two parity circuits and their coherent combination, the good amplitudes are $\tilde{q}_j/(2s\sqrt{N})$. Write $\tilde{\alpha} = \sum_j |\tilde{q}_j|^2$. Choose an integer r and scale

$$p_r = \sin^2 \frac{\pi}{4r+2}, \quad s = \sqrt{\frac{\tilde{\alpha}}{4Np_r}}, \quad (15)$$

with s large enough to keep both parity polynomials below the prescribed bound. Standard Grover iterations then produce $(-1)^r \tilde{q}/\sqrt{\tilde{\alpha}}$ with unit ideal success. The conjugate preparation followed by its inverse supplies the second factor $\tilde{q}_j$, yielding $\sum_j \tilde{q}_j^2 V_j / \tilde{\alpha}$. All Grover reflections and both

preparations occur in the executed circuits. Their workspace can be reused because amplification uncomputes it; standalone QuCF checks quantify the residual numerical error. The known factor $\tilde{\alpha}$, rather than a fitted scale or phase, recovers the physical unnormalized state.

For polar compilation, substitute $\tilde{q}_j = \tilde{a}_j e^{i\phi_j}$ in Eq. (13). Exact diagonal phase lookup leaves $|\tilde{q}_j - q_j| = |\tilde{a}_j - a_j|$; phase-gate roundoff is checked by the complex standalone circuit. The single even-polynomial preparation has amplitudes $i^d \tilde{a}_j / (s\sqrt{N})$ rather than the Cartesian LCU's $\tilde{q}_j / (2s\sqrt{N})$, so use $s = \sqrt{\tilde{\alpha}/(Np_r)}$. The known real global phase $i^d(-1)^r$ cancels between preparation and its inverse in Eq. (6). Both diagonal layers contain the same phase $e^{i\phi_j}$, supplying $e^{2i\phi_j}$, not its modulus. The cost of this exact finite-node phase representation is $O(N)$ controlled phase and X gates. No polynomial-in- n_k claim is made for that representation.

B Additional circuit and calibration details

The selected kernel parameters, both QSVT polynomial degrees, Grover round counts and known one-norms are given in Tables 4 and 5. The core and (6, 9) baseline exponent is $\beta = 0.7$; the added (6, 6) values are explicitly listed. The original search grid for f_2 was $c \in \{0.2, 0.25, 0.3, 0.35, 0.4, 0.5, 0.75, 1\}$, $\gamma \in \{2, 3, 4, 5, 6, 8, 10, 12\}$ and integer $K \in [4, 60]$. For each core configuration the score was

$$\mathcal{C} = 2(2r_0 + 1)(d_R + d_I)(n_k + 6) + 8Kt\|L\|_2(2n_k + 4N_x), \quad (16)$$

where r_0 is the preliminary rounded Grover estimate from the sampled peak of the square-root weights and a polynomial amplitude scale of 0.8. This is only a declared calibration proxy; all reported costs come from the later compiler and emulator. The final exact-round normalization can increase this preliminary round count. The candidate's modal error had to be below 0.6ϵ , and the variant additionally below 0.98 times the selected baseline error. Its ideal success had to exceed both 0.21 and 1.002 times the baseline value. When the coarse baseline had no feasible point, the minimum-error baseline was retained. The f_2 candidates at those points were selected from the target-feasible, success-feasible set. The full candidate grid and first, less restrictive pre-validation selection are retained. No final gate count was inferred from this score.

The added required-register study first scanned $\beta = 0.7$ at six dyadic times $t = 0.1/2^j$, $j = 0, \dots, 5$, and $K = 4, 4.25, \dots, 60$. Its diagnostic candidates are preserved, but this initial stage was not separately timed. The refinement uses $\beta = 0.7, 0.705, \dots, 0.8$ and adds $t = 0.2, 0.4, 0.8$. It evaluated 42 525 candidates in 15.86 seconds. Bounded Nelder–Mead refinements and the associated variant grids evaluated 44 502 candidates in 15.81 seconds. The final variant grid evaluated 14 400 candidates in 5.18 seconds. These are distinct recorded stages, including repeated parameter evaluations, not independent circuit trials. The final baseline is the minimum-proxy grid point meeting the relaxed 0.95ϵ guard and 0.21 ideal-success floor. The variant also meets error < 0.98 times baseline and success ≥ 1.002 times baseline. At the infeasible $t = 0.1$ control, minimum-error choices replace the target constraints as specified in the frozen plan. No new cutoff, kernel parameter or time is selected after the Cartesian or polar executions.

The symmetric phase solver initializes all independent phase offsets to zero around endpoint phases $\pi/4$. A Newton step uses the explicit product-rule Jacobian of the two-dimensional QSP transfer matrix and backtracking when needed. The convergence tolerance is 2×10^{-11} on its collocation grid. Each sequence is independently evaluated on 4097 points and rejected if its maximum polynomial discrepancy exceeds 10^{-8} . Boundedness is checked on 16385 points with the derivative remainder in Eq. (14). No external phase-finding code is used.

Table 4: Kernel parameters and Cartesian QSVT degrees. Each row is a distinct weight circuit; repeated times reuse angles.

Family	(n_k, ϵ)	K	Parameters	Degree	r	$\tilde{\alpha}$
β	$(5, 1 \times 10^{-2})$	11	$\beta = 0.7$	30/31	6	1.254469
f_2	$(5, 1 \times 10^{-2})$	6	$(c, \gamma) = (0.4, 3)$	30/31	5	1.200247
β	$(6, 3 \times 10^{-3})$	22	$\beta = 0.7$	62/63	8	1.295161
f_2	$(6, 3 \times 10^{-3})$	12	$(c, \gamma) = (0.3, 5)$	62/63	7	1.194715
β	$(6, 1 \times 10^{-3})$	17.75	$\beta = 0.765$	62/63	7	1.427630

Family	(n_k, ϵ)	K	Parameters	Degree	r	$\tilde{\alpha}$
β	$(6, 1 \times 10^{-3})$	19.75	$\beta = 0.75$	62/63	7	1.395088
β	$(6, 1 \times 10^{-3})$	20.75	$\beta = 0.745$	62/63	7	1.384731
β	$(6, 1 \times 10^{-3})$	21.5	$\beta = 0.7$	62/63	8	1.294913
f_2	$(6, 1 \times 10^{-3})$	6.75	$(c, \gamma) = (0.5, 3)$	62/63	6	1.332639
f_2	$(6, 1 \times 10^{-3})$	11	$(c, \gamma) = (0.25, 5)$	62/63	7	1.134861
β	$(7, 3 \times 10^{-2})$	11	$\beta = 0.7$	80/127	6	1.257263
f_2	$(7, 3 \times 10^{-2})$	5	$(c, \gamma) = (0.35, 3)$	126/125	5	1.129490
β	$(7, 1 \times 10^{-2})$	18	$\beta = 0.7$	126/125	7	1.291248
f_2	$(7, 1 \times 10^{-2})$	8	$(c, \gamma) = (0.4, 4)$	112/127	6	1.272199
β	$(7, 3 \times 10^{-3})$	22	$\beta = 0.7$	126/127	8	1.297597
f_2	$(7, 3 \times 10^{-3})$	13	$(c, \gamma) = (0.25, 6)$	124/127	7	1.159320
β	$(7, 1 \times 10^{-3})$	29	$\beta = 0.7$	126/127	9	1.302132
β	$(7, 1 \times 10^{-3})$	32	$\beta = 0.7$	124/127	9	1.302892
f_2	$(7, 1 \times 10^{-3})$	17	$(c, \gamma) = (0.35, 6)$	126/127	8	1.284828
f_2	$(7, 1 \times 10^{-3})$	18	$(c, \gamma) = (0.35, 6)$	126/127	9	1.285176
β	$(9, 1 \times 10^{-3})$	32	$\beta = 0.7$	508/509	9	1.303120
f_2	$(9, 1 \times 10^{-3})$	18	$(c, \gamma) = (0.35, 6)$	508/279	9	1.285179

Table 5: Kernel parameters and Polar QSVT degrees. Each row is a distinct weight circuit; repeated times reuse angles.

Family	(n_k, ϵ)	K	Parameters	Degree	r	$\tilde{\alpha}$
β	$(6, 1 \times 10^{-3})$	17.75	$\beta = 0.765$	62	3	1.427630
β	$(6, 1 \times 10^{-3})$	19.75	$\beta = 0.75$	62	4	1.395088
β	$(6, 1 \times 10^{-3})$	20.75	$\beta = 0.745$	62	4	1.384731
β	$(6, 1 \times 10^{-3})$	21.5	$\beta = 0.7$	62	4	1.294913
f_2	$(6, 1 \times 10^{-3})$	6.75	$(c, \gamma) = (0.5, 3)$	62	3	1.332639
f_2	$(6, 1 \times 10^{-3})$	11	$(c, \gamma) = (0.25, 5)$	62	3	1.134861

B.1 Additional executed comparisons

The reference direct-weight compiler prepares $\sqrt{w_j}$ with its native controlled rotations and uses the rounded integer Grover count. If $\alpha = \sum_j |w_j|$, its single-preparation success amplitude is

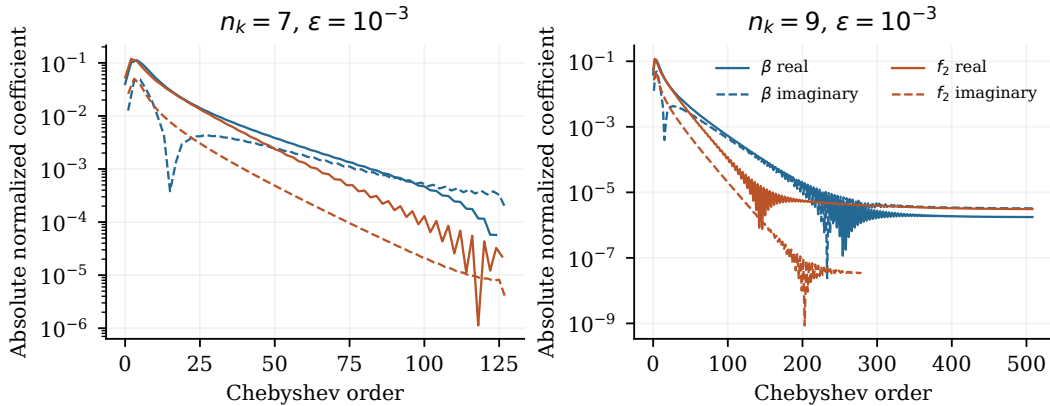


Figure 6: Normalized Chebyshev coefficients actually retained in the original core weight polynomials (left) and their larger-register counterparts (right). Only the permitted parity orders are shown. The real parts remain close to the interpolation degree limit; the approximate kernel’s imaginary component truncates earlier in the larger register. Endpoint behavior and discrete aliasing, in addition to real-axis kernel decay, therefore affect the weight circuit’s cost.

Table 6: Complete required-register study, including both QSVT compilers and the native direct control at every time. All five $t = 0.1$ circuits fail the 10^{-3} target, as marked by $\dagger$. Parameters were frozen before execution; the polar compiler was declared after the Cartesian runs without retuning kernels.

Method	Test	(n_x, n_k, t, ϵ)	Rel. error	STMC	Q_A	Anc.	p_{succ}
NJ direct	ADE	$(6, 6, 0.003125, 1 \times 10^{-3})$	8.93×10^{-4}	15 332	552	16	0.4788
β polar	ADE	$(6, 6, 0.003125, 1 \times 10^{-3})$	8.93×10^{-4}	23 718	552	15	0.4847
f_2 polar	ADE	$(6, 6, 0.003125, 1 \times 10^{-3})$	2.90×10^{-4}	20 356	416	15	0.5566
β Cart.	ADE	$(6, 6, 0.003125, 1 \times 10^{-3})$	8.93×10^{-4}	55 739	552	16	0.4847
f_2 Cart.	ADE	$(6, 6, 0.003125, 1 \times 10^{-3})$	2.90×10^{-4}	46 771	416	16	0.5566
NJ direct	ADE	$(6, 6, 0.00625, 1 \times 10^{-3})$	9.24×10^{-4}	31 375	1 200	16	0.4983
β polar	ADE	$(6, 6, 0.00625, 1 \times 10^{-3})$	9.24×10^{-4}	42 545	1 200	15	0.5018
f_2 polar	ADE	$(6, 6, 0.00625, 1 \times 10^{-3})$	3.46×10^{-4}	23 722	552	15	0.5499
β Cart.	ADE	$(6, 6, 0.00625, 1 \times 10^{-3})$	9.24×10^{-4}	71 782	1 200	16	0.5018
f_2 Cart.	ADE	$(6, 6, 0.00625, 1 \times 10^{-3})$	3.46×10^{-4}	50 137	552	16	0.5499
NJ direct	ADE	$(6, 6, 0.0125, 1 \times 10^{-3})$	9.19×10^{-4}	49 982	1 952	16	0.4947
β polar	ADE	$(6, 6, 0.0125, 1 \times 10^{-3})$	9.19×10^{-4}	61 152	1 952	15	0.4975
f_2 polar	ADE	$(6, 6, 0.0125, 1 \times 10^{-3})$	7.77×10^{-4}	37 389	1 104	15	0.5372
β Cart.	ADE	$(6, 6, 0.0125, 1 \times 10^{-3})$	9.19×10^{-4}	90 389	1 952	16	0.4975
f_2 Cart.	ADE	$(6, 6, 0.0125, 1 \times 10^{-3})$	7.77×10^{-4}	63 804	1 104	16	0.5372
NJ direct	ADE	$(6, 6, 0.1, 1 \times 10^{-3})$	$3.59 \times 10^{-3\dagger}$	355 709	14 304	16	0.4424
β polar	ADE	$(6, 6, 0.1, 1 \times 10^{-3})$	$3.59 \times 10^{-3\dagger}$	366 879	14 304	15	0.4424
f_2 polar	ADE	$(6, 6, 0.1, 1 \times 10^{-3})$	$1.44 \times 10^{-3\dagger}$	187 071	7 152	15	0.5787
β Cart.	ADE	$(6, 6, 0.1, 1 \times 10^{-3})$	$3.59 \times 10^{-3\dagger}$	401 718	14 304	16	0.4424
f_2 Cart.	ADE	$(6, 6, 0.1, 1 \times 10^{-3})$	$1.44 \times 10^{-3\dagger}$	219 094	7 152	16	0.5787

Table 7: Core time sweep at fixed precision target. Kernel parameters were selected by the declared calibration rule.

Method	Test	(n_x, n_k, t, ϵ)	Rel. error	STMC	Q_A	Anc.	p_{succ}
β Cart.	ADE	$(4, 7, 0.05, 1 \times 10^{-3})$	5.66×10^{-4}	138 828	1 104	17	0.5110
f_2 Cart.	ADE	$(4, 7, 0.05, 1 \times 10^{-3})$	4.04×10^{-4}	128 119	552	17	0.5251
β Cart.	ADE	$(4, 7, 0.1, 1 \times 10^{-3})$	5.22×10^{-4}	153 623	1 800	17	0.4543
f_2 Cart.	ADE	$(4, 7, 0.1, 1 \times 10^{-3})$	4.40×10^{-4}	139 854	1 104	17	0.4670
β Cart.	ADE	$(4, 7, 0.2, 1 \times 10^{-3})$	5.83×10^{-4}	180 819	3 032	17	0.3793
f_2 Cart.	ADE	$(4, 7, 0.2, 1 \times 10^{-3})$	5.24×10^{-4}	145 619	1 952	17	0.3892

$b = \sin[(2r + 1) \arcsin \sqrt{\alpha/N}]$. For the matched direct controls, multiplying the raw output by α/b^2 restores the physical state. This known normalization is recorded in each manifest. The success probability still includes the actual b^4 loss, and all rounds are counted. The unchanged archival audit additionally retains the source input-preparation sampler; its output is compared directly with the archived output, without mixing its counts with the new matched comparisons.

Table 8: Matched direct-weight controls. Their states are rescaled by the analytically known Grover success amplitude, not by fitting the exact solution.

Method	Test	(n_x, n_k, t, ϵ)	Rel. error	STMC	Q_A	Anc.	p_{succ}
β direct	ADE	$(4, 7, 0.1, 3 \times 10^{-2})$	1.25×10^{-2}	21 841	832	17	0.4816
β direct	ADE	$(4, 7, 0.1, 1 \times 10^{-2})$	5.32×10^{-3}	27 621	1 104	17	0.4566
β direct	ADE	$(4, 7, 0.1, 3 \times 10^{-3})$	1.70×10^{-3}	39 356	1 656	17	0.4551
β direct	ADE	$(4, 7, 0.1, 1 \times 10^{-3})$	5.22×10^{-4}	42 416	1 800	17	0.4516
β direct	ADE	$(4, 5, 0.1, 1 \times 10^{-2})$	$1.31 \times 10^{-2\dagger}$	17 429	832	15	0.4574
β direct	ADE	$(4, 6, 0.1, 3 \times 10^{-3})$	$3.89 \times 10^{-3\dagger}$	35 994	1 656	16	0.4567
β direct	ADE	$(4, 7, 0.05, 1 \times 10^{-3})$	5.66×10^{-4}	27 621	1 104	17	0.5079
β direct	ADE	$(4, 7, 0.2, 1 \times 10^{-3})$	5.83×10^{-4}	68 586	3 032	17	0.3770

C Artifact map and reproduction

All outputs are under `proposal/`. The primary plans are `code/kernel_plan.json`, `code/kernel_plan_selected.json`, and `code/kernel_plan_implementation.json`. Their timestamps precede validation. The separately timestamped `code/kernel_plan_extension.json` declares the later larger-register and direct-weight tests. The required-register calibration stages are `kernel_plan_headline.json`, `kernel_plan_headline_refinement.json`, `kernel_plan_headline_local_refinement.json`, and `kernel_plan_headline_final_design.json`, all under `code/`. The frozen eight selections are in `code/kernel_plan_headline_selected.json`. `code/kernel_plan_polar.json` records the subsequent compiler change before its runs. Historical plans were preserved. The implementation record is `codebase_notes.md`; the complete development and rejected attempt record is `attempts_log.md`.

Each directory under `logs/runs/`, `logs/polar_runs/` or `logs/direct_controls/` contains its oracle, QSP/QSVT phase files, execution timestamp and command, raw QuCF HDF5 output, flat circuit, state arrays and result JSON. The exact state is computed by `scipy.linalg.expm` from the constructed generator during that run. `logs/weights/` and `logs/polar_weights/` retain the coefficients, phase iterations, verification grids, native angle files, standalone raw preparation and amplified preparation executions. The unchanged-source audit is in `logs/reference_reproduction/`. The files `logs/all_results.csv`, `logs/gate_tallies.json`, `logs/analysis.json` and `logs/number_traceability.json` map the manuscript’s values to these sources. Tables, numeric macros and figures are generated, not transcribed manually.

The source patch adds resource metadata only; it does not alter any gate’s action. A copied generator oracle marks one existing leaf gate, and its marker survives copying, controls and adjoints. Summing the markers gives Q_H, Q_L . As a separate structural check, their total equals four times the number of executed qubitization operators in these selector circuits. The flat gate lists independently sum to QuCF’s native gate count. Source hashes and build/environment logs are retained. The HDF5 dependency and a compatible NumPy installation are local to the artifact.

`code/reproduce.sh` rebuilds the emulator, independently checks calibration against the frozen selection, generates missing phases, executes the circuit suites, and regenerates the paper. Existing completed runs are reused by the drivers; a clean output directory is required for independent timing measurements. The raw-output audit in `code/audit_artifact.py` recomputes each exact reference, checks every reported metric against its HDF5 source, checks oracle and phase-file hashes, and verifies that no planned configuration is omitted. The executed experiment suite is deterministic. All figure data are available without extracting values from the PDF.